higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

COMMUNICATION-ELECTRONICS N5

15 NOVEMBER 2019

This marking guideline consists of 6 pages.

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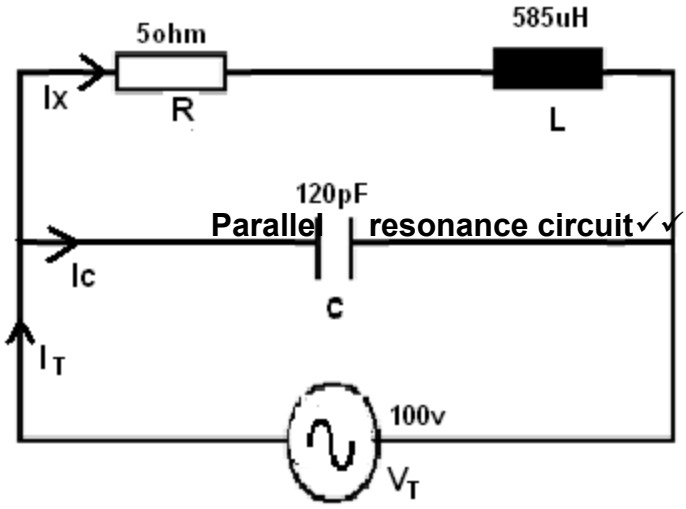
QUESTION 1

1.1 D
1.2 F
1.3 E
1.4 B
1.5 K
1.6 H
1.7 G
1.8 C
1.9 I
1.10 A

(10 × 1) **[10]**

QUESTION 2: AC NETWORKS

2.1 2.1.1



(2)

2.1.2 $Zd = \frac{L}{CR}$

$= \frac{585 \quad x \quad 10^{-6}}{120 \quad x \quad 10^{-12} \quad x \quad 5}$ ✓

= 975 KΩ✓ (2)

2.1.3 $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$

$= \frac{1}{5}\sqrt{\frac{585 \quad x \quad 10^{-6}}{120 \quad x \quad 10^{-12}}}$ ✓

= 442✓ (2)

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2.1.4 Ic = IQ

But $I = \frac{V}{Zd}$

$= \frac{100}{975 \times 10^3}$ ✓

= 102,56 µA✓

∴ Ic = $102,56 \times 10^{-6} \times 442$✓

= 45 mA✓ (4)

2.1.5 $Fr = \frac{1}{2\pi\sqrt{LC}}$

$= \frac{1}{2\pi\sqrt{585 \ x10^{-6} \ x \ 120 \ x \ 10^{-12}}}$✓

= 601 kHz✓ (2)

2.2 $Q = \frac{XL}{R}$

And XL = 2 πfL

= 2 π × 601×10^3x 585×10^{-6}✓

= 2209,36 Ω✓ (2)

∴ $Q = \frac{XL}{R}$

$= \frac{2209,36}{5}$✓

= 441, 81 = 442✓ (2)

2.3 The co-efficient of coupling is defined as the degree of coupling between the primary and secondary windings and also it is the ratio of mutual inductance to maximum inductance. (4)

2.1 2.4.1 (a) Primary
(b) Impedance
(c) Coupled
(d) Reflected (4)

2.4.2 (a) Voltage
(b) Current
(c) Lags (3)

2.4.3 (a) Secondary
(b) Induced
(c) Winding (3)

[30]

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QUESTION 3: FOUR TERMINAL NETWORKS

3.1 $Zo = \sqrt{Zoc \ . Zsc}$

$Zoc = Z1 + Z2 = 10 + 40$✓

$= 50\ \Omega$✓ (2)

$Zsc = Z1 + \frac{Z1 \ x \ Z2}{Z1 + Z2}$

$= 10 + \frac{10 \ x \ 40}{10 + 40} = 10 + 8$✓

$= 18\ \Omega$✓ (2)

$Zo = \sqrt{Zoc \ . Zsc}$

$= \sqrt{50 \ x \ 18}$✓

$= 30\ \Omega$✓ (2)

3.2 $$\text{Attenuation in dB} = 10 \log \frac{P_1}{P_2}$$

$$= 10 \log \frac{3}{5 \times 10^{-3}}$$ ✓

$= 27,78$ dB✓ (2)

Since 1 Neper = 8,686 dB

$\therefore$ the Neper = $\frac{27,78}{8,686}$ ✓ = 3,2 Neper ✓ (2)

[10]

QUESTION 4: FILTERS

4.1.1 (a) T – type High pass filter✓

(b) π – type High pass filter✓ (2)

4.1.2 $$FC = \frac{1}{4\pi\sqrt{LC}}$$

$$= \frac{1}{4\pi\sqrt{0,13 \ x 10^{-6} x \ 4,77 x 10^{-3}}}$$

$$= \frac{1}{4\pi \ X \ 0,000007874}$$

$= 10104,276$Hz $= 10,1$ kHZ✓ (3)

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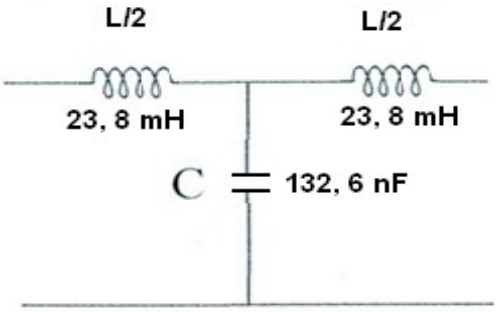
4.2 (a) $C = \frac{1}{\pi \times Zo \times fc} = \frac{1}{\pi \times 600 \times 4 \times 10^3}$ ✓

= 132, 6 nF✓ (2)

(b) $L = \frac{Zo}{\pi \times fc} = \frac{600}{\pi \times 4 \times 10^3}$ ✓

= 47, 7 mH✓ (2)

(c)



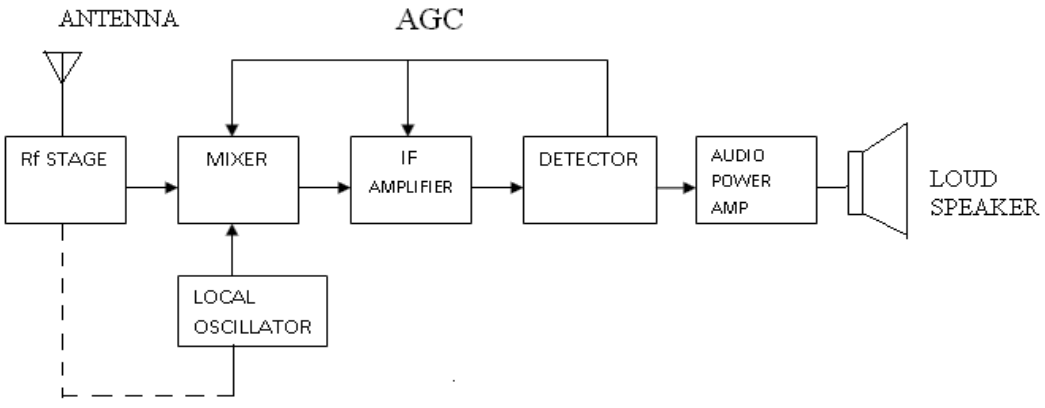
T-type Low Pass filter✓ (1)
[10]

QUESTION 5: RECEIVERS AND TRANSMITTERS

5.1 5.1.1 It simplifies the equipment of transmitter and receiver.
5.1.2 It uses a simple envelope detector in the receiver.
5.1.3 It reduces the cost of the receiver.

(3 × 2) (6)

5.2



(5)

5.3 AM Superheterodyne receiver.✓✓✓✓✓

ANTENNA: Converts electromagnetic radio waves into electrical Rf currents.✓
RF STAGE: Uses tuned circuits to select the desired RF carrier signal. Provides selectivity to the receiver.✓
MIXER: Combines the incoming Rf with the frequency of the local oscillator and beats the two frequencies to produce a third frequency (IF) which is equal to the difference between the two frequencies and is called the intermediate frequency.✓
OSCILLATOR: Produces a RF of constant amplitude and at a fixed difference frequency above or below the incoming RF. The oscillator frequency is applied to the mixer for producing the IF.✓
IF AMPLIFIER: Provides the main amplification to the receiver and gives a high degree of stability to the IF signal. It also provides additional selectivity.✓
DETECTOR: Recovers the original modulation signal from the carrier and eliminates the carrier. Provides the AGC voltage.✓
AUDIO AMPLIFIER: Amplifiers the audio signal to a suitable level to drive the loud speaker✓
LOUDSPEAKER: Converts electrical audio frequency signal into mechanical sound waves.✓
AGC: Varies the overall gain of a radio receiver automatically with the changing strength of the received signal.✓ (9 × 2) (18)
[29]

QUESTION 6: ANTENNAE

6.1 6.1.1 The mismatch in space (1)

6.1.2 The tips of the transmission lines are close together so cancellation occurs. (2)

6.2 It is the power received by an antenna which is directly proportional to the collecting area of the antenna. (2)
[5]

QUESTION 7: BODE PLOTS

7.1 Bode diagrams are an approximation of the frequency and phase response of an RC circuit. (3)

7.2 They are used to improve the accuracy of bandwidths and also provide amplifier stability. (3)
[6]

TOTAL: 100

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